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The cost of post quantum cryptography on drone and robot microcontrollers: evidence, models and a migration path

Author: Huda Mueen. Draft of 2026-10-21. Generated from `docs/PAPER_body.md` by `scripts/build_paper.py`; every number below is read from a file in this repository, and `scripts/check_numbers.py` fails the build when a stated number drifts from its file. Licence: all rights reserved (`LICENSE`).

Target venue: a six page workshop paper with one additional page (PPT-PQC 2027, `data/raw/S13_venue.md`). Sections 1 to 6 and the references form the submission; the section marked Supplementary is for the repository version only. Length of the main part (abstract to conclusion, table included, figure captions and references excluded): 4035 words, about 4.5 pages at 900 words per two column page, before the references and the three figures, each about a third of a column high.

Abstract

Drones and ground robots authenticate commands, telemetry and firmware with classical public key cryptography on microcontroller flight computers and narrow radio links. This paper asks what the standardised replacements (ML-KEM, ML-DSA, LMS) cost on such devices and links, how well that cost can be predicted before a board is measured, and which devices in a fleet can migrate in place. It combines 1,515 published rows from 60 source URLs, the newest each read against its source in two separate reading passes, with a trace driven cycle model, energy, link, boot, fleet and mission models, and checks of those models by independent tools: memory of real flight stack builds, the ns-3 network simulator, the ProVerif protocol verifier and cycle accurate simulation of two open cores, plus software in the loop runs of a flight stack. A cycle model fitted on half of the published rows predicts the other half of the same code base with a median error of 1.0 percent, but 9.8 percent for a scheme left out of the fit. Measured flight stack memory raises the estimated median share of fleet devices that need hardware replacement from 0.186 to 0.398 under stack optimised thresholds and from 0.201 to 0.300 under thresholds drawn from published rows, a correction of our earlier estimate. In ns-3, an IEEE 802.15.4 transfer that the link model puts at 7.1 s reaches the published 30 to 60 s, on average, with any one of three configurations the source does not state, so the published gap has sufficient causes but no proven one. A protocol verifier checked 57 queries, including deliberate controls built to fail, proved 39, and found three design flaws whose fixes the reference architecture adopts. On telemetry class radios a hybrid key establishment followed by a symmetric session keeps command availability where it is without authentication, while a post quantum signature on every command does not. Nothing in this study was measured on a microcontroller; a board session that can refute the cycle model is prepared and its outcome registered in advance.

1. Introduction

A small multirotor or ground robot runs its flight stack on a microcontroller and talks to its operator over a telemetry radio, a mesh network or a cellular link. Public key cryptography protects its command channel, its update channel and sometimes its data. A future quantum computer threatens these differently: recorded data is exposed today if it stays valuable longer than the migration takes; command authentication is exposed from the day forgery becomes possible; and a boot chain whose root of trust is fixed in read only memory cannot be fixed at all.

The replacements are standardised: ML-KEM for key establishment [nist_nistpubs_fips_nist_fips_203], ML-DSA and SLH-DSA for signatures [nist_nistpubs_fips_nist_fips_204] [nist_nistpubs_fips_nist_fips_205], and stateful hash based schemes for firmware [rfc-editor_rfc_rfc8391]. What an operator cannot yet say from evidence is what they cost on its own flight controller, over its own link, across its own fleet. Published microcontroller measurements are many but uneven: most come from one core, from libraries tuned for a benchmark framework [iacr_2019_844], under clock, cache and counter settings that differ from paper to paper [iacr_2026_1333].

The study answers six research questions. RQ1: what the algorithms cost in cycles and memory on microcontroller cores, and how well cycles can be predicted without a board. RQ2: what they cost in energy and CPU headroom. RQ3: how handshake, command and update traffic behave over the radios such vehicles use. RQ4: what signed boot and update cost. RQ5: which devices of a fleet built from real board classes can migrate in place. RQ6: which command and update architecture is feasible.

Contributions. (1) A provenance controlled evidence table whose rows were checked against their sources by readers who worked separately from the staging and from each other (every row of the last two rounds by two readers), with measured reading error rates. (2) Models whose errors are measured on held out data, including leave one group out tests that expose where they do not transfer. (3) Checks of those models by independent tools: real flight stack builds, a network simulator, a protocol verifier and two open core simulations. (4) A reference architecture for an authenticated command and update channel, with three design changes found by formal verification. (5) A pre registered board experiment that can refute the cycle model.

2. Related work

The pqm4 framework set the common method for Cortex-M4 measurements [iacr_2019_844] [githubusercontent_mupq_pqm4_master_benchmarks_csv]; later work moved optimised code to the Cortex-M7 [iacr_2025_366], reduced memory for ML-DSA [iacr_2022_323], and measured small cores such as the Cortex-M0+ [arxiv_pdf_2603_19340]. A recent methodology paper shows that results move with library configuration, caches and statistics and proposes reporting rules [iacr_2026_1333]. LMS and XMSS have been compared on the Cortex-M4 [iacr_2020_470], post quantum secure boot has been built and measured [date-conference_proceedings_archive_2020_pdf_0438], open bootloaders document post quantum options [githubusercontent_wolfssl_wolfboot_master_docs_pq_md], and post quantum signed updates over constrained networks have been measured [iacr_2021_781]. MAVLink 2 signs messages with a shared secret [mavlink_en_guide_message_signing]; post quantum command authentication for unmanned systems has been studied under real time limits [iacr_2026_2343], and constrained nodes have been integrated with ROS 2 security [ipp_docs_secure_integration_of_extremely_resource_constraine]. This paper adds the combination: one verified table across these strands, models validated against it and against independent tools, and fleet and mission consequences derived from it, each with its uncertainty and evidence label.

3. Method

Evidence labels. Every number carries one label, never mixed in one column: published (copied from a retrieved source), host measured, emulated, estimated (arithmetic from published values), modelled (with its validation error), RTL simulated (cycle accurate simulation of an open core

design), simulated (network) (ns-3 output) and planned. Board measured is reserved for a session on a real board.

Evidence table. Rows enter `data/evidence.csv` only through staging files and a merge script that changes no value. In this round a systematic search staged 265 rows (stream S13); two readers who saw neither the staged values nor each other's sheet read every row, 0 value fields disagreed with staging, all 41 wording disagreements got a third look, and 187 rows were promoted (`docs/VERIFICATION6.md`); of the others, 47 were demoted after review because they broke a written inclusion or exclusion rule (`data/raw/S13_demoted.csv`) and 31 had a value field that only one reader confirmed. The 393 wording disagreements that round 4 had left without a third look were all read again by blind readers; the two rounds used different rules, so their rates are given apart: in the round 4 sample, where any difference from the source counted, staging was wrong in 6 of 60 (10 percent, Wilson interval 4.7 to 20.1); in this round, where only a contradiction of the source counted, staged text contradicted the source in 1 of 365 settled lines (Wilson interval 0.05 to 1.5 percent), although the staged text differed verbatim from the source in 153 of them; 28 lines stay ambiguous (`docs/STATISTICS_V5.md`).

Models and independent checks. A trace driven cycle model counts executed instructions by class in the pinned pqm4 code and maps them to cycles with costs fitted on half of the matched published rows. An energy model multiplies cycles by datasheet currents; a link and protocol simulator uses cited radio parameters with Monte Carlo intervals; a boot model adds hashing and verification; a fleet planner classifies devices of nine real board classes against published memory footprints; a mission simulation compares four security architectures. Each was checked this round by a tool that does not share its code: the memory of real flight stack builds for the planner, ns-3 for the link model, ProVerif for the protocol design, and a second open core for the RTL results. Statistics use a cluster bootstrap by binary and leave one group out validation; one chain propagates published ranges end to end with Sobol indices.

Pre registration. Predictions with refutation clauses were hashed before each round and are scored by script; misses stay in the record (`docs/PREDICTIONS_V5.md`, `docs/PREDICTIONS_V5_BOARD.md`). Scores of this round are in `docs/PREDICTIONS_V5_SCORES.md` and stay partial until the round closes.

4. Results

Each answer states its uncertainty and its evidence label. Detailed results per study are in the Supplementary section and the study documents.

4.1 RQ1: compute cost and its prediction

Answer. Within the pqm4 code base, cycles can be predicted without a board to within a few percent, but the fitted model does not transfer to a scheme it has not seen, and published figures for one operation spread mainly with the statistic and compiler they report. Fitted on half of the matched published rows and tested on the other half (split by binary), the cycle model's median absolute error is 1.0 percent; a cluster bootstrap by binary gives 0.8 to 1.7 percent, almost the round 4 interval, but the board prediction intervals widen by a factor of 1.68 when computed by cluster. Leaving out a whole implementation family gives 1.8 percent, and leaving out a whole scheme 9.8 percent (cluster interval 6.8 to 11.8, worst 29.9); a reduced four feature model transfers better (2.9 percent). Figure 1 shows all five designs. Signing remains the hardest operation because its cost varies from message to message. Evidence: modelled, from emulated instruction traces and published cycle counts.

A hierarchical model of the published cycle counts (crossed random effects for implementation variant, statistic, compiler and source) attributes 39.4 percent of the within cell variance to the statistic and 49.9 percent to the compiler, but both intervals are wide and the fit is driven by the few rows that report a worst observed signing value: without them, the residual standard deviation falls from 0.38 to 0.21 in log cycles and the variant share rises from 3.6 to 32.7 percent.

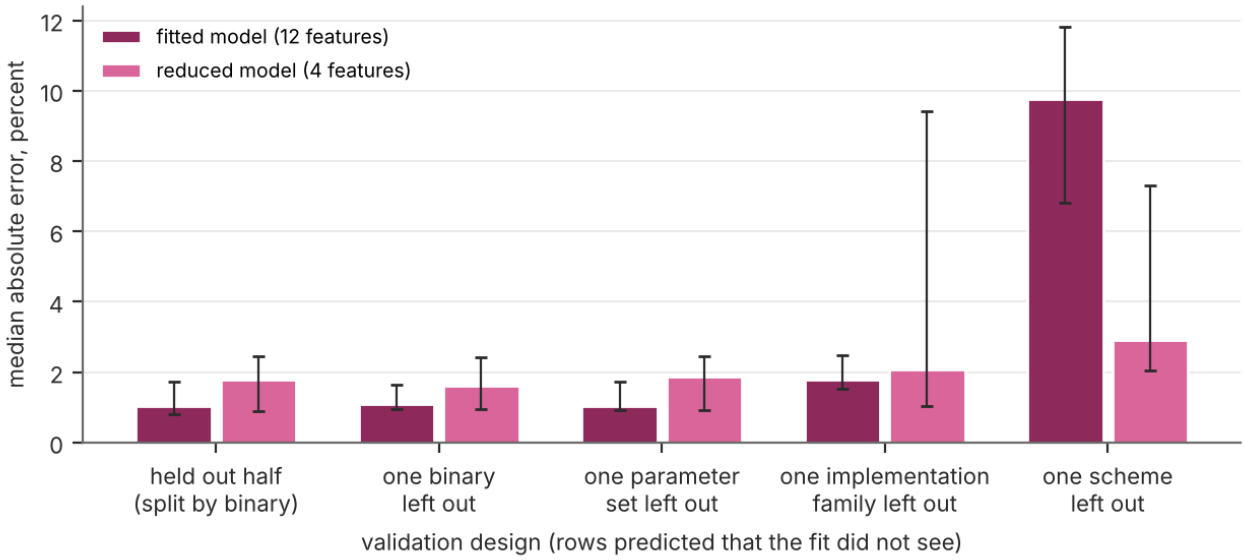


Figure 1: Median absolute error of the cycle model on rows the fit did not see, by validation design, with cluster bootstrap 95 percent intervals (modelled). Source: `stats/results/cycle_cluster_bootstrap.csv` and `stats/results/cycle_lofo_summary.csv`.

Both fits are reported; neither is the single truth. As a reference that no vendor setting can change, the portable C code of mlkem-native and mldsa-native ran on cycle accurate simulations of two open RV32IMC cores (RTL simulated): ML-KEM-768 decapsulation takes 3,033,704 cycles on Ibex and 2,070,034 on CV32E40P, which is the faster core in 18 of 18 operations, by a factor of 1.33 to 1.47. These are simulations of core designs with memory without wait states, not measurements of silicon.

4.2 RQ2: energy and CPU headroom

Answer. CPU time, not battery energy, is the binding budget. At 20 percent CPU headroom a 168 MHz Cortex-M4 flight controller can verify about 23.6 ML-DSA-44 signatures per second against about 1,377 messages under a symmetric MAC session (estimated). Verifying a 1 MB firmware image costs a median of 29.8 mJ (2.5 to 97.5 percentile 7.4 to 81.6) in the end to end chain, and the datasheet current per MHz carries most of that range (total Sobol index 0.58). Calibrated on one board and one source, published energy is 2.21 times the uncalibrated estimate, so absolute energies are likely low by a similar factor; rates depend only on cycles. Evidence: estimated and modelled.

4.3 RQ3: links and protocols

Answer. On telemetry class radios a hybrid handshake costs seconds once per session; on IEEE 802.15.4 meshes, transfer times depend on settings that published figures do not state, and our link model is optimistic there. The protocol simulation (1000 seeded trials per cell) gives a median time to the first authenticated command of 1.39 s for the MAVLink style hybrid profile on a telemetry radio without loss, 1.24 s on one 802.15.4 hop and 1,162 s on LoRaWAN DR5, where the duty cycle dominates (modelled). For a 50 kB transfer over one 802.15.4 hop with ns-3 3.48 defaults, ns-3 gives 5.72 s (simulated (network)) and linksim 7.12 s (modelled), against a published 30 to 60 s. Three changes each bring the ns-3 mean into that range (Figure 2): a minimum and maximum backoff exponent of 8, allowed by the cited standard (40.1 s); 3 hops (55.2 s); or 16 neighbours sending, where only the mean reaches the range and its interval extends below it (33.2 s). The source states none of these settings, so they are sufficient causes in simulation, not

the proven cause. ns-3 also shows that linksim's store and forward over hops is optimistic: at 3 hops linksim gives 21.4 s against 55.2 s in ns-3, because frames of different hops contend, which linksim does not model. Of 241 ns-3 runs, 23 did not complete and are excluded from the verdict. The selective repeat check against the closed form 1/(1-P), on its first registered run, placed 4 of 5 loss rates inside their 95 percent interval; the miss at P = 0.3 did not recur in ten follow up seed blocks added afterwards and labelled as such.

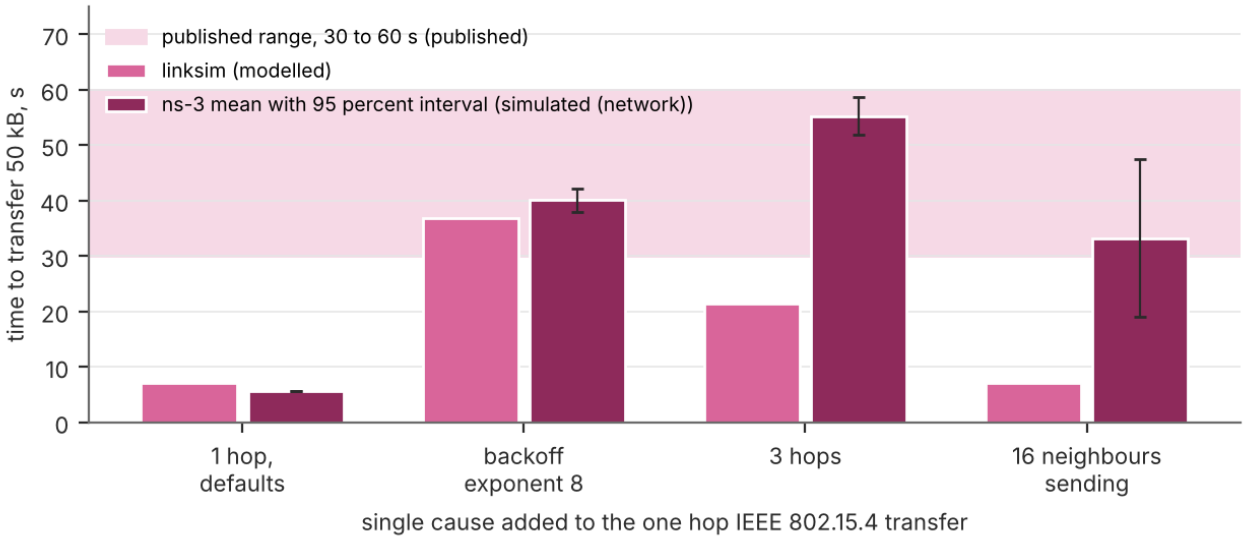


Figure 2: Time to transfer 50 kB over IEEE 802.15.4 for one hop with defaults and for each single cause that reaches the published range: linksim (modelled) beside ns-3 (simulated (network), mean and 95 percent interval), with the published range shaded. Source: `net-sim/results/gap_sweep.csv`.

4.4 RQ4: signed boot and update

Answer. For firmware, the signature scheme matters less than image size and hash hardware, and a version counter is required. For a 1 MB image on a 168 MHz Cortex-M4 with software SHA-256, verification takes about 462 ms with ML-DSA-44 and 470 ms with LMS, of which hashing is 454 ms (estimated). Across the end to end ranges the median is 456 ms (340 to 609), ranked by clock (total Sobol index 0.53) and hash cost per byte (0.44); the implementation factor of the signature does not register. Against published bootloader timings the calibrated check rests on three rows, so these are estimates of order. Formal verification (Section 4.6) found that a signature without a signed version and a monotonic counter accepts a downgrade (F3). A fixed first stage that verifies only a classical signature cannot be migrated; such devices need hardware replacement for signing migration.

4.5 RQ5: fleet migration

Answer, a correction of round 4. With measured flight stack memory, a larger share of devices needs hardware replacement than round 4 reported. Default builds of PX4 v1.17.0 and ArduPilot Copter-4.7.2 succeeded for 10 targets on 7 board classes (host measured from linker output); PX4 leaves as little as 6,008 B of flash free. With these values in place of the round 4 assumption, the median share needing hardware replacement rises from 0.186 to 0.398 under stack optimised thresholds and from 0.201 to 0.300 (bootstrap interval of the median 0.297 to 0.303; it describes the median, not the spread of the share across draws) under thresholds drawn from published rows (Figure 3). Round 4's statement that the share stays near one fifth no longer holds. Using the larger of the two stacks' free memory per board, the uncertain scenario gives 0.265. Free

RAM is an upper bound because heap use at run time is not visible in a build. Evidence: estimated, from host measured memory.

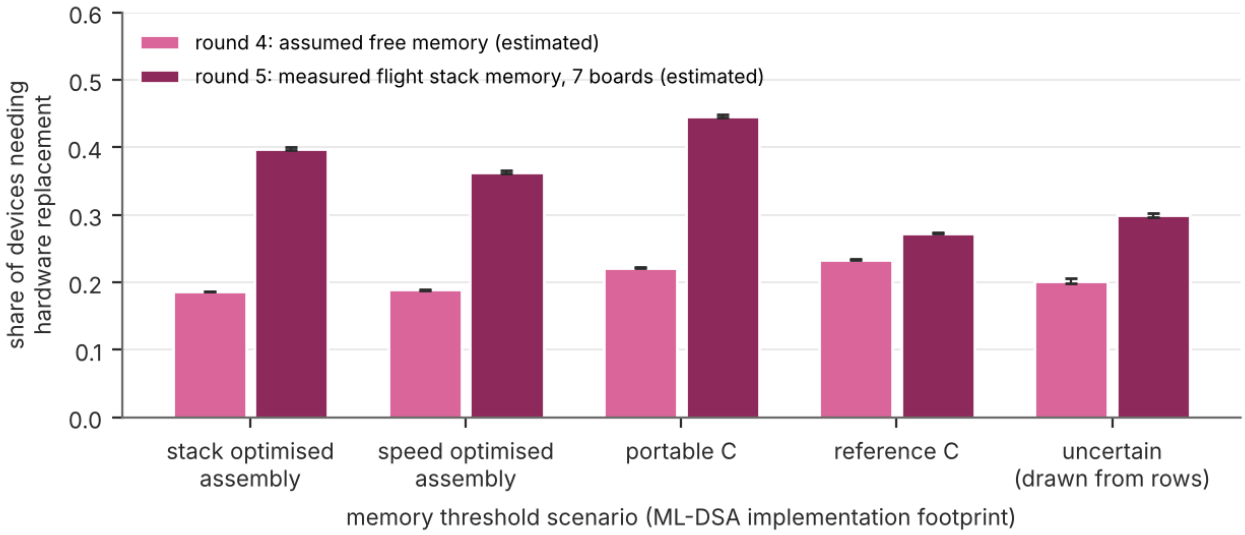


Figure 3: Share of fleet devices that need hardware replacement, median over 200 draws with a bootstrap interval of the median, under the round 4 assumed free memory and with measured flight stack memory for seven boards (estimated). Source: `planner/results/flightstack_mc_summary.csv`.

4.6 RQ6: a feasible architecture

Answer. Hash based signatures (LMS) with a signed version and a monotonic counter for firmware; a hybrid key establishment in which both sides sign, followed by a symmetric MAC session, for commands on telemetry class links or faster; and no per command signatures on streamed commands. In the mission simulation (1000 runs per configuration) the hybrid session keeps the drone's command link available about as often as an unauthenticated stream on a telemetry radio (mean 0.991 against 0.992 without authentication, at a 0.5 s deadline), while a post quantum signature on every command adds about 380 ms per command and lowers availability to 0.794 (estimated). Across the end to end ranges the hybrid availability has median 0.946 (0.798 to 0.999), and the outage rate and mean outage length rank first and second (total Sobol indices 0.55 and 0.48). Within the assumed outage ranges and with message sizes held fixed, outages rather than cryptography dominate the variation of availability; the ranking reflects those assumed ranges. Evidence: estimated (mission) and modelled (end to end chain).

ProVerif 2.05 checked 57 queries in 17 models of the stated designs, the changed designs and deliberate controls (Table 1). Three findings changed the design and were adopted in the reference architecture: in the stated session only the ground station signs, so the vehicle is not authenticated to it (F1; fixed by a vehicle ML-DSA-44 signature over the transcript and key confirmation, which adds a fourth flight to the handshake); the provisioned key fallback accepts a recorded command again after a power cycle (F2; fixed by deriving a session key from fresh nonces); and firmware without a signed version and counter accepts a downgrade (F3). The fallback's lack of forward secrecy is inherent and documented. These are results under the Dolev Yao model with abstract primitives, not statements about an implementation. Evidence: modelled.

Table 1. ProVerif results by outcome (modelled). Source: `formal/results/proverif_results.csv`.

Outcome	Queries	Meaning
proved	39	property holds for every run the model allows: unbounded sessions for the command channel; the firmware model is bounded to three releases and three boots per device
counterexample	17	attack trace found: stated designs (F1 to F3) and deliberate controls with broken components
cannot be proved	1	MAVLink timestamp rule with gaps; its weaker non injective form holds

Board experiment. A board kit builds images for an STM32L4R5 (Cortex-M4) and an STM32L552 (Cortex-M33) board, now with an option that switches the caches off; 36 cycle predictions for the caches off pass were registered before any session, all out of sample (planned). The option is checked only by disassembly of the images and, for the L552, an emulator that logs the register write but does not model a cache, and the Cortex-M33 rows apply Cortex-M4 costs, an unsourced transfer assumption.

4.7 Software in the loop

ArduPilot Copter 4.7.2 ran in software in the loop with its built in vehicle model, behind a relay that emulates a SiK class link with frame loss and one outage per mission, and a ground station script that sent a stream of acknowledged commands (`docs/SITL_STUDY.md`). PX4 v1.17.0 was not used because its MAVLink module has no message signing to serve as the classical baseline. Each of four configurations flew 20 scripted missions with stored seeds. With the link up, MAVLink 2 signing as the flight stack ships it added 3.7 ms to the median command round trip and the hybrid session with design change F1 3.9 ms (SITL measured). The one way added latency of the hybrid session, 1.94 ms (emulated by the relay with the mission study's own inputs, so a consistency check of the relay, not a test of the model), equals the mission study's 1.95 ms. The emulated hybrid handshake of four flights took 1.28 s against 1.42 s when the mission handshake model is applied to the flight sizes and compute of this configuration (the mission study's own first session time is 1.45 s). With outages, the share of commands acknowledged within 0.5 s was 0.914 for the hybrid session against 0.916 in the mission study at the nearest point of its outage rate sweep; one outage per short flight is a higher rate than that point, and the share of flight time in outage differs between configurations because each draws its own outages, so differences between configurations of a few hundredths are not resolved. The relay of the hybrid configuration enforced no replay rule, so this figure is not the availability of the recommended design with its sequence rule. A post quantum signature on every command added 382 ms one way (emulated), as the mission study predicts (380 ms), but 496 ms on the round trip to the acknowledgement, and only 0.419 of its commands were acknowledged within 0.5 s, against 0.730 in the mission study, which counts one way delivery: the deadline falls in the tail of a delivery of several hundred milliseconds. In the configuration with MAVLink 2 signing as shipped, the flight stack discarded 16 delivered signed commands, because the emulated link layer's retransmission let a frame arrive after a newer one and fail the signing timestamp rule, an effect the mission study does not model. The strict next sequence rule that the formal proofs use would reject such frames too, and every frame after a lost one, so a deployment needs in order delivery below the session layer or a replay window that the proofs do not yet cover; this is an open design question, not a result in favour of either rule. Configurations 3 and 4 add the bytes and published compute delays on the link rather than executing cryptography, and the link is an emulation, not a radio. Evidence: SITL measured for round trips through the flight stack, which is the part that tests the

model; emulated for one way and handshake figures; against the mission study (estimated).

5. Threats to validity

1. Nothing was measured on a microcontroller. Every cycle, energy and timing figure is published, emulated, RTL simulated, simulated (network), estimated or modelled.
2. The board kit's caches off option is untested on silicon: the emulator logs the register write but does not model a cache, and no free emulator models the STM32L4R5.
3. The Cortex-M33 board predictions apply Cortex-M4 instruction costs, a transfer assumption without a source.
4. The cycle model is fitted and tested within one code base; it does not transfer to an unseen scheme (Figure 1), and its external check on another library and core rests on few rows.
5. The hierarchical meta analysis is driven by the few rows that report a worst observed value; statistic and compiler are confounded with source and have wide intervals, so both fits are reported.
6. Flight logs could not be used: the PX4 log sites disallow automated access in their robots rules and no reuse licence was found, so mission traffic is synthetic.
7. Free RAM after a flight stack build is an upper bound, since heap use at run time is not visible; fleet counts are synthetic, and one board definition states more flash than the reference profile.
8. The ns-3 causes of the 802.15.4 gap are sufficient in simulation, not proven: the published figure states no hop count, load or backoff setting. Hop count, spacing and background traffic are assumptions, and ns-3 is a simulator, not a radio.
9. The committed ns-3 runs used the earlier three flight handshake; the round 3 link study and its design notes still describe three flights. The mission, protocol, energy and end to end studies use the four flight handshake of finding F1.
10. Energy rests on datasheet currents and one calibration source; the end to end chain ran 20 missions per evaluation, not 1000.
11. Formal results hold under the Dolev Yao model with abstract perfect primitives, one identity per side, without the rekey grace period or the LMS signer state; they say nothing about an implementation.
12. The match rules for the third look at wording disagreements were written after a first literal comparison was seen; ambiguous lines stay unsettled and are listed for the author.
13. IEEE Xplore and the ACM Digital Library could not be searched (their search pages could not be fetched), and Semantic Scholar refused; the systematic search under covers those indexes.
14. The selective repeat follow up blocks and the extra ns-3 runs were added after the first results; predictions are scored on the first, registered runs only.
15. Software in the loop used one flight stack (ArduPilot), one link profile (SiK class) and one outage per mission; the hybrid session and the per command signature were emulated by their bytes and published compute delays, not executed, and a first set of runs with a timing defect was replaced by a rerun with the same seeds.

6. Conclusion

On the links small drones and ground robots use, the cost of post quantum authentication is a one time handshake if the design uses a session; on telemetry radios and 802.15.4, per command signatures are the option that fails, and on LoRaWAN no command stream fits the duty cycle. For firmware, hashing dominates unless the device has hash hardware, and a signed version with a counter is required. The fleet answer is set by how much memory the flight stack leaves free, and measuring it raised the replacement share well above the round 4 estimate. A cycle model predicts within a code base but not across schemes, so migration decisions should rest on a measurement of the deployed code; the registered board session is the next step and can refute the model.

References

Keys refer to `data/bibliography.bib`, generated from the evidence table and the source notes.

Supplementary material (repository version only; not part of the six page submission)

S1. Detailed results by research question

S1.1 RQ1: compute cost, and how well it can be predicted Evidence. The table holds 1,515 rows; 812 are microcontroller rows with a cycle count, on 8 core labels as the sources print them (Cortex-M4 and Cortex-M4F are counted apart, and one label is 'not stated'). Inside groups of the same parameter set, operation and core class that have rows from two or more sources, the largest published value is 1.44 times the smallest at the median group (up to 4.67; ratios computed from published rows; `docs/META_ANALYSIS.md`). Part of this spread is deliberate: a group can hold speed and stack optimised variants of one scheme and both means and medians, so it measures how far a published figure can sit from the deployed code when the variant and conditions are not known, not the measurement error of one implementation. Library, optimisation, toolchain, cache and statistic all move it.

Prediction without a board. A trace driven model counts executed instructions by class in the pinned pqm4 code (emulated) and maps them to cycles (modelled; `docs/CYCLE_MODEL.md`). Fitted on half of the matched published rows and tested on the other half, split by binary so that no test row shares a binary with a training row, its median absolute error is 1.0 percent (95 percent interval 0.7 to 1.7, a row bootstrap; the cluster bootstrap of round 5 gives almost the same interval, section 4.1) on 27 held out rows, and 0.8 percent without signing; signing misses by up to 22.8 percent because its cost varies from message to message. The two halves share algorithms and much code (for example the same parameter set in two assembly variants), so this is a test of interpolation within one code base, not of transfer to new code. Without fitting, a model built from assumed per instruction costs is 12.6 percent high at the median. Applied without refitting to an independent low memory library, key generation lands 2.9 to 8.3 percent above its published Cortex-M33 values, a weak check because core, memory system and compiler all differ. The model's predictions for the first board session are registered with intervals in `docs/PREDICTIONS_V4_BOARD.md`.

Cycle accurate simulation of an open source core. The portable C code of mlkem-native and mldsa-native was run on a cycle accurate simulation of the open source Ibex RV32IMC core (small configuration, memory without wait states; RTL simulated, `docs/RTL_SIM_RESULTS.md`): ML-KEM-768 decaps 3,033,704 cycles; ML-KEM-512 decaps 1,915,739 cycles; ML-DSA-44 verify 5,405,374 cycles; ML-DSA-44 sign 12,553,829 cycles (signing is the median of five messages and varies several fold). These are measurements of a core design in simulation, not of Arm silicon. They are 1.24 to 2.99 times the published portable C rows for the Cortex-M4 (different library, compiler, instruction set and statistic, so only the order of magnitude is compared) and give a microcontroller class reference that no vendor setting can change.

S1.2 RQ2: energy and CPU headroom Energy per operation was estimated from published cycles and cited run currents of five parts and checked against published energy rows: uncalibrated, the estimate is low by a median of 54.8 percent on the held out rows; after one scale factor fitted on the other half, the median error is 0.7 percent (`docs/ENERGY_STUDY.md`). The check uses one board and one source, so it calibrates a factor rather than validating the model in general, and the energy figures elsewhere in this paper are uncalibrated; on that board the published energy is 2.21 times the estimate, so they are likely low by a similar factor; the message rates below depend only on cycles and are unaffected. At 20 percent CPU headroom a 168 MHz Cortex-M4 flight controller can verify about 23.6 ML-DSA-44 signatures per second, 46.5 Ed25519 signatures and 12.6 LMS signatures, against about 1,377 messages under a symmetric MAC session (estimated). Energy per hour is small for every design; CPU time, not battery, is the binding budget, and the command rate is the input that matters most (Sobol indices in `energy/results/sobol.csv`).

S1.3 RQ3: links and protocols The link model was repaired one cited cause at a time and checked against published figures and closed forms (`docs/LINKSIM_STUDY.md` section 4.4). The LoRa time on air formula reproduces the published examples, but this is a weak check: only one of the four examples states every parameter, and for the other three the unstated settings were chosen after the fact because they reproduce the published value. Fragment counts agree with an independent count and the duty cycle wait equals its formula, an arithmetic check; stop and wait retransmissions in the event simulation agree with their closed form within Monte Carlo intervals; selective repeat, which the protocol and mission studies use, was checked against its closed form in round 5, with the outcome in section 4.3. For IEEE 802.15.4, adding every cited cost (PHY header, 6LoWPAN fragment headers and alignment, acknowledgements, inter frame spacing, clear channel assessment and backoff, turnaround) takes a 50 kB transfer from 4.5 s to 7.1 s, which is still 4.2 times faster than the published 30 to 60 s. Most of that gap is unexplained by single hop costs in the link model; the round 5 ns-3 runs (section 4.3) find settings that close it, none of which the source states, so the model is reported as optimistic on 802.15.4 networks. The rate part was checked against one published table of handshakes over an emulated link, with one constant fitted on two of its rows; that is a weak test on one testbed.

The protocol simulation (1000 seeded trials per cell, `docs/PROTOCOL_SIM.md`) gives the time to the first authenticated message. On a telemetry radio and on 802.15.4, a hybrid session handshake completes in about one to two seconds; on LoRaWAN it takes minutes at DR5 and hours at DR0, because of the duty cycle. Per message post quantum signatures overload the slower links.

S1.4 RQ4: signed boot and update For a 1 MB image on a 168 MHz Cortex-M4 (the STM32F405 class of a small drone flight controller) with software SHA-256, verification takes about 462 ms with ML-DSA-44 and 470 ms with LMS, of which hashing the image is 454 ms. With a hash peripheral, which the STM32F405 itself lacks (the rows are a what if for a part of that class with one), the totals fall to 23 ms and 30 ms; the signature then becomes the larger part for the slower verifiers, while for Ed25519, ML-DSA-44 and ML-DSA-65 hashing still costs about as much as the signature or more. The choice of signature scheme changes boot time far less than image size and hash hardware do (`docs/BOOT_UPDATE_STUDY.md`). Against published bootloader timings the uncalibrated model is far off and the calibrated check rests on three rows, so these are estimates of order, not predictions. A fixed first stage root of trust that verifies only a classical signature cannot be changed by any of this; such devices need hardware replacement for signing migration.

S1.5 RQ5: fleet migration The round 3 synthetic fleet is reproduced exactly (0.260 to 0.382 of devices need hardware replacement). On fleets drawn from nine real board classes, with every memory threshold drawn from published rows, the median share is 0.201 (2.5 to 97.5 percentile 0.186 to 0.232). These round 4 figures assumed the free memory a flight stack leaves. Round 5 measured it for seven board classes from real flight stack builds, and the replacement share rose (section 4.5 and `docs/FLIGHTSTACK_STUDY.md`); the round 4 statement that the share is close to 0.20 no longer holds. Under the Mosca inequality with stated ranges, data lifetime has the largest total Sobol index (0.91), largely because its range is the widest. Policy dates come from primary texts (`docs/POLICY_TIMELINE.md`); the supplied UAE policy texts give no post quantum date.

S1.6 RQ6: mission level design In the mission simulation (1000 runs per configuration, `docs/MISSION_STUDY.md`), a classical session, a hybrid session and hash signed firmware with session keys keep the drone's control link available about as often as an unauthenticated stream on a telemetry radio (mean over runs 0.991 for the hybrid session; estimated): within the assumed outage ranges, link outages rather than authentication set availability. A post quantum signature on every command adds about 380 ms per command on that radio (estimated), lowers availability to 0.794, and overloads the link for a ground robot streaming commands at 5 Hz (utilisation 1.88). LoRaWAN carries none of the command streams within its duty cycle. The resulting reference architecture (`docs/REFERENCE_ARCHITECTURE.md`): hash based signatures (LMS)

for firmware, a hybrid key establishment followed by a symmetric MAC session for commands on telemetry class links or faster, and no per command signatures on streamed commands.

S2. Discussion

Three results change what an engineer should do. First, a published cycle count is not a property of the algorithm alone: the implementation variant and the measurement conditions move the same operation on the same core class, so a migration decision should rest on a measurement of the deployed code, which the board kit makes cheap. Second, on the links drones and robots use, the cost of post quantum authentication is a one time handshake, not a per message burden, provided the design uses a session: on telemetry radios and 802.15.4, per message signatures are the option that fails, and on LoRaWAN no command stream fits the duty cycle. Third, the fleet answer depends on how much memory the flight stack leaves free; measured builds show less free flash than round 4 assumed, and operators can settle this from their own build output.

Where the models overlap they are not independent: the mission model reads the link simulator's parameters through the shared inputs file, so their agreement on a single handshake is a consistency check, not a validation. The energy and mission studies both find CPU time, not energy, to be the binding budget. ns-3 now gives sufficient causes for the 802.15.4 gap and shows that the link model's multi hop assumption is optimistic; the link model was not tuned to the published figure.

S3. Reproducibility: one command per figure and table

Each figure and table regenerates from committed scripts with stored seeds; third party code is fetched at pinned versions and never committed. Heavy steps (flight stack builds, ns-3, open core simulation) need the pinned toolchains listed in each study document.

Item	Command	Result file
Figure 1 (cycle model errors)	`python3 stats/cycle_validation.py && python3 scripts/make_paper_figures.py`	`stats/results/cycle_cluster_bootstrap.` `stats/results/cycle_lofo_summary.csv`
Figure 2 (802.15.4 transfer by cause)	`bash netsim/fetch_build.sh && python3 -m netsim.run_netsim && python3 -m netsim.make_study && python3 scripts/make_paper_figures.py`	`netsim/results/gap_sweep.csv`
Figure 3 (fleet replacement share)	`python3 planner/flightstack_study.py && python3 scripts/make_paper_figures.py`	`planner/results/flightstack_mc_summary`
Table 1 (ProVerif results)	`sh formal/install.sh && python3 -m formal.build_models && python3 -m formal.run_all`	`formal/results/proverif_results.csv`
Flight stack memory (RQ5)	`bash flightstack/fetch.sh && bash flightstack/build_all.sh && python3 flightstack/extract_memory.py`	`data/flightstack_memory.csv`
Hierarchical meta analysis (RQ1)	`python3 stats/meta_hierarchical.py`	`data/meta/hierarchical_variance.csv`

Item	Command	Result file
End to end chain and Sobol ranking	`python3 stats/end_to_end.py`	`stats/results/e2e_mc_summary.csv`, `stats/results/e2e_sobol.csv`
Open core cycles (RQ1)	`bash rtlsim/run.sh && bash rtlsim/run_cv32e40p.sh && python3 rtlsim/report.py`	`rtlsim/results/core_spread.csv`
Board predictions, caches off	`python3 bench/board_kit/predictions_v5.py`	`bench/board_kit/predictions_v5_board.c`
This paper and the claims register	`python3 scripts/build_paper.py && python3 scripts/check_numbers.py`	`data/claims_register.csv`
PDF	`./scripts/build_pdfs.sh`	`docs/PAPER.pdf`

S4. Earlier verification and review

Earlier rounds verified rows by separate reading passes inside this project on random and headline samples (details in `docs/VERIFICATION.md`, `docs/VERIFICATION2.md` and `docs/VERIFICATION3.md`) and rechecked 403 rows from supplied papers in full; in round 4 every new row was read by two readers (8 of 439 staged rows with values had a wrong value before correction, 1.8 percent, Wilson interval 0.9 to 3.6 percent; `docs/VERIFICATION5.md`). Predictions of earlier rounds: `docs/PREDICTIONS.md`, `docs/PREDICTIONS_V2.md`, `docs/PREDICTIONS_V3.md`, `docs/PREDICTIONS_V4.md`. Each round was reviewed by fresh reviewers working from a clean copy of the repository only; they belong to this project, not to an outside body, and no external reviewer has checked the study. Findings and resolutions are logged in `docs/REVIEW_FINDINGS*.csv`.